

Wenxian Shi

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Education

Massachusetts Institute of Technology

Cambridge, MA

Sep. 2021-present

- Ph.D. in Computer Science.
- Advised by Prof. Regina Barzilay.

Peking University

Beijing, China

Sep. 2016-Jul. 2019

- M.S. in Computer Science.
- Advised by Prof. Junfeng Hu.

Peking University

Beijing, China

Sep. 2012-Jul. 2016

- B.S. in Physics.

Professional

Genesis Therapeutics

California, USA

Jun. 2024-Aug. 2024

- Machine Learning Research Intern.
- Advised by Zach Kagin and Maruan Al-Shedivat.

Bytedance AI Lab

Shanghai, China

Jul. 2019-Aug. 2021

- Researcher in Machine Learning and Natural Language Processing.
- Advised by Prof. Lei Li and Prof. Hao Zhou.

Research Interests

My research interests lie in generative models for discrete and structured data, including text generation and the generation of biological sequences such as proteins.

Publications (* for Equal Contribution)

- Shi, W., Wu, M., & Barzilay, R. (2025). Predicting sub-population specific viral evolution. In *Transactions on Machine Learning Research (TMLR)*.
- Shi, W., Wohlwend, J., Wu, M., & Barzilay, R. (2023). Improving influenza A vaccine strain selection through deep evolutionary models. *bioRxiv*.
- Song, Z., Zhao, Y., Shi, W., Jin, W., Yang, Y., & Li, L. (2023). Generative Enzyme Design Guided by Functionally Important Sites and Small-Molecule Substrates. In *Forty-first International Conference on Machine Learning (ICML)*.
- Shi, W.*, Song, Y.*, Zhou, H., Li, B., & Li, L. (2021). Follow your path: a progressive method for knowledge distillation. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*.
- Shi, W., Zhou, H., Miao, N., & Li, L. (2020, November). Dispersed exponential family mixture VAEs for interpretable text generation. In *International Conference on Machine Learning (ICML)*.
- Ye, R., Shi, W., Zhou, H., Wei, Z., & Li, L. (2019, September). Variational Template Machine for Data-to-Text Generation. In *International Conference on Learning Representations (ICLR)*.
- Miao, N., Zhou, H., Zhao, C., Shi, W., & Li, L. (2019). Kernelized Bayesian softmax for text generation. *Advances in Neural Information Processing Systems (NeurIPS)*.
- Shi, W. X., & Chen, Q. B. (2015). Wobbling geometry in a simple triaxial rotor. *Chinese Physics C (CPC)*.

Selected Projects

Improving Influenza A vaccine strain selection through deep evolutionary models

MIT CSAIL, advised by Regina Barzilay

Jan. 2023-Present

- Discover a computational metric that quantifies the antigenic similarity of vaccine strains with future dominant strains, which demonstrates a strong correlation with clinical vaccine effectiveness.
- Introduce a deep learning-based approach that predicts both the antigenic properties of vaccine strains and the dominance of future circulating viruses, enabling efficient virtual screening of a large number of vaccine compositions.
- Utilize historical antigenic analysis data from the WHO to demonstrate that our model selects vaccine strains that reliably improve over the recommended ones.

Knowledge distillation via exploring local targets

ByteDance AI-Lab, advised by Hao Zhou and Lei Li

2021

- Propose a simple yet effective knowledge distillation method that projects the supervision signals of a teacher model into the student's parameter space.
- Outperform strong knowledge distillation baselines and achieve state-of-the-art performance in various image and text classification tasks.

Dispersed Exponential Family Mixture VAEs for interpretable text generation.

ByteDance AI-Lab, advised by Hao Zhou and Lei Li

2020

- Introduce VAEs with a mixture of exponential family distribution for interpretable text generation.
- Theoretically analyze the general mode-collapse problem in a mixture of exponential family VAEs and propose an effective method to fix it.
- Experimental results show that our approach does induce a more meaningful space, and it outperforms strong baselines in text generation and dialog generation.